

SURYANSH KUMAR

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EDUCATION

VIT Bhopal University

B.Tech, Computer Science and Engineering (Core)

Bhopal, MP

2023 – 2027

Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, Software Engineering

TECHNICAL SKILLS

Languages: Java, Python, C++, SQL (MySQL), JavaScript

Backend & Web: Node.js, Express.js, Flask, React, REST APIs, HTML, CSS

Databases: PostgreSQL, MySQL, MongoDB, SQLite

AI / ML: FAISS, LangChain, Ollama, scikit-learn, Pandas, NumPy, Retrieval-Augmented Generation

Tools & Platforms: Git/GitHub, AWS (EC2, Lambda), Linux, VS Code, Jupyter Notebook

EXPERIENCE

DRDO — Summer Trainee (Returning)

Apr 2026 – Jun 2026

Multimodal RAG Document Assistant | [GitHub](#)

Python, Flask, Ollama (Llama 3, LLaVA, nomic-embed-text), FAISS, LangChain, PyMuPDF, pdfplumber, SQLite, REST APIs

- Built an offline multimodal RAG assistant in Python and Flask that answers natural-language questions over uploaded PDFs, grounding every answer in the source document and citing exact page numbers to reduce hallucination.
- Engineered a multimodal ingestion pipeline extracting text (PyMuPDF), tables (pdfplumber) and per-page image descriptions from a local vision model, extending question answering to charts, tables and scanned figures.
- Orchestrated 3 locally hosted models through Ollama — Llama 3 for generation, nomic-embed-text for embeddings, LLaVA for vision — against a per-document FAISS index, keeping all data on-device and removing recurring cloud API cost.
- Implemented session-based authentication, role-based access control and an admin dashboard on SQLite, enabling multi-user access and independent addition or removal of document indexes.

DRDO — Winter Trainee

Nov 2025 – Jan 2026

OSINT Investigation Platform | [GitHub](#)

Node.js, Express.js, React, PostgreSQL, Prisma ORM, Python, Ollama (Llama 3), REST APIs

- Engineered a full-stack OSINT platform in Node.js, Express and React that automates entity extraction from documents and file metadata, replacing manual triage in investigative workflows.
- Built an iterative investigation pipeline that extracts entities from uploaded documents and integrates a locally hosted Llama 3 model via Ollama for keyword extraction and report generation.
- Designed a normalized PostgreSQL schema across 6+ core entities with Prisma ORM, improving data integrity and enabling relationship mapping across cases.
- Wrote 5+ custom OSINT automation scripts to collect and enrich intelligence from public web sources.

PROJECTS

Medication Recommender System using Machine Learning

Python, scikit-learn, Pandas, NumPy

- Built an end-to-end ML classification system using Decision Tree, SVM and KNN algorithms trained on 3,000+ symptom records for symptom-to-medication mapping.
- Engineered 15+ features from raw medical datasets with Pandas and NumPy, improving model accuracy ~25% over the untuned baseline.
- Compared 3 models under 5-fold cross-validation, selecting the best performer at ~89% held-out accuracy.

Mind Connect — Mental Health Support Platform

React.js, Node.js, Express.js, JWT, bcrypt

- Architected a full-stack web application supporting anonymous posting and real-time community interaction.
- Implemented JWT-based authentication and bcrypt (10-round hashing) for secure session and access management.
- Designed a normalized schema behind a stateless REST API layer while maintaining data-privacy compliance.

Automatic Change Detection Using Ultrasonic Radar

Arduino, C++, HC-SR04 Ultrasonic Sensors, Signal Processing

- Engineered an Arduino-based ultrasonic radar using HC-SR04 sensors for real-time change detection across a 180° scanning range, with distance accuracy within ± 3 cm.
- Applied FFT and low-pass filtering to clean sensor data, reducing noise interference ~40% and improving measurement reliability.

CERTIFICATIONS

- AWS Technical Essentials — EC2, Lambda, S3, IAM; The Bits and Bytes of Computer Networking — Coursera; NPTEL, IIT Madras.